

Warren Kozak

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EDUCATION

Carleton College

September 2023 – June 2027

Bachelor of Arts in Computer Science, Mathematics; Cumulative GPA: 3.9/4.0

Northfield, MN

- Relevant Coursework: Discrete Math, Advanced Software Design, Programming Languages, Introduction to Computer Systems, Robotics, Parallel and Distributed Systems, Multivariable Calculus, Linear Algebra, Ordinary Differential Equations, Computational Linear Algebra

RESEARCH EXPERIENCE

Northwestern University (TIILT Lab)

July 2025 – August 2025

NSF REU Undergraduate Researcher

Evanston, IL

- Prototyped wearable devices for use in swimming using a Seeed Studio nRF52840 Sense
- Optimized data logging with FreeRTOS, increasing IMU sample rate from 25 to 50 Hz for more reliable data
- Utilized a Mahony filter algorithm to generate quaternions from 6-axis IMU data to detect device orientation
- Developed an LSTM model to classify IMU data into stroke categories, achieving an accuracy of 95%
- Presented data logging and LSTM classification findings at the 2025 Carleton College Undergraduate Research Symposium

INDUSTRY EXPERIENCE

Garmin

June 2026 – August 2026

Embedded Software Engineering Intern

Olathe, KS

- Incoming Embedded Software Engineering Intern for Summer 2026 on GNSS Platforms Team

Hildebrand Technology

June 2024 – August 2024

Summer Intern

London, UK

- Developed a Python-based Home Assistant integration utilizing MQTT and REST APIs to process real-time IoT energy data for Hildebrand Glow devices

SELECTED PROJECTS

Multi-Agent Robotic Pursuit & Evasion | ROS2, Gazebo, Python

April 2026 – Present

- Simulating multi-agent robotic pursuit and evasion with ROS2 and Gazebo with artificial potential fields
- Writing a short technical report highlighting our approach and findings

N-Body Physics Sim | C++, Raylib, AGAMA

March 2026 – Present

- Implementing an n-body physics simulator with raylib graphics
- Utilizing Barnes-Hut algorithm to increase simulation efficiency together with Verlet integration

T5-Harmony Music Generation | Python, T5-Base, PyTorch

September – November 2025

- Fine-tuned a seq2seq T5-Base LLM on 30,000+ songs to understand and generate symbolic music
- Evaluated model output using quantitative metrics and human evaluation
- Authored a formal technical report detailing training approach and results

TECHNICAL SKILLS

Languages: C++, C, Python, Lisp/Scheme, $\LaTeX$

Robotics & Embedded: ROS2, FreeRTOS, Arduino, PlatformIO, MQTT

Simulation & ML Libraries: Gazebo, TensorFlow/Keras, AGAMA, NumPy, pandas

Developer Tools: UNIX/Linux, Git/GitHub, Docker, Neovim

LEADERSHIP & ACTIVITIES

CS Course Staff | Carleton College

September 2025 – Present

- Evaluating and providing technical feedback for student submitted work in Discrete Math and Algorithms

Team Captain & Academic All-District | Men's Varsity Swim & Dive

September 2023 – Present

- Leading daily captain's practices during the off-season, helping coaches coordinate and lead team
- Swam at 2025 MIAC Championships, achieving all-MIAC in the 200 medley and freestyle relays